Lab 14: GANs

Intro to Deep Learning, Spring 2026

Outline

1. **Lecture Recap:** What GANs are and how they work

2. **Why GANs Are Hard to Train:** Instability, balance, and convergence issues

3. **How GANs Are Stabilized:** Key ideas such as loss, distance, and constraints

4. **What to Look for in the Notebook:** How to read the training loop and outputs

5. **GAN Notebook**

LECTURE RECAP

The problem

- From a large collection of images of faces, can a network learn to *generate* new portrait
  - Generate samples from the distribution of “face” images
    - How do we even characterize this distribution?

LECTURE RECAP

Discriminative vs Generative Models

Given a distribution of inputs X and labels Y.

Discriminative models

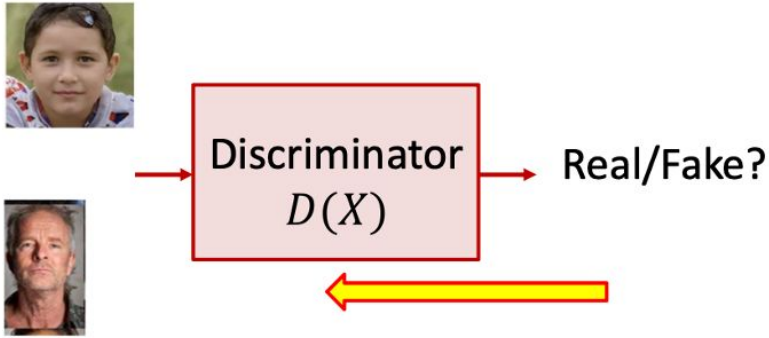
- Discriminative models learn conditional distribution P(Y | X)
- Learns decision boundary between classes.
- Limited scope. Can only be used for classification tasks.
- E.g. Logistic regression, SVM etc.

Generative models

- Generative models learn the Generative models learn the joint distribution P(Y, X)
- Learns actual probability distribution of data.
- Can do both generative and discriminative tasks.
- E.g. Naïve Bayes, Gaussian Mixture Model etc.
- Harder problem, requires a deeper understanding of the distribution than discriminative models.

LECTURE RECAP

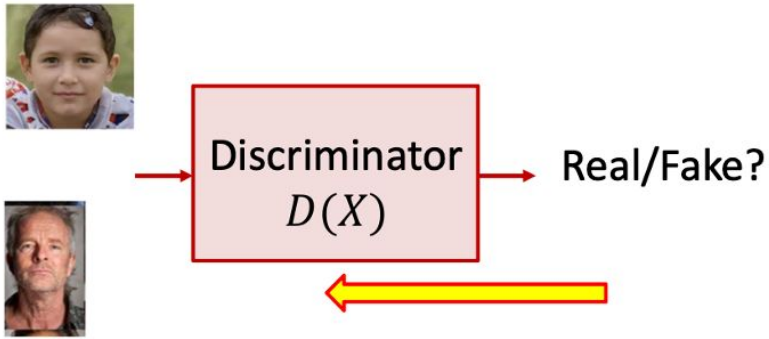
Training the discriminator



- **Training the discriminator:**
  - The discriminator is provided training examples of real and synthetic faces
  - The discriminator is trained to minimize its classification loss
    - Minimize error between actual and predicted labels
  - Discriminator parameters are trained such that
    - $D(X) = 1$ for real faces
    - $D(X) = 0$ for synthetic faces (i.e $1 - D(X) = 1$)

LECTURE RECAP

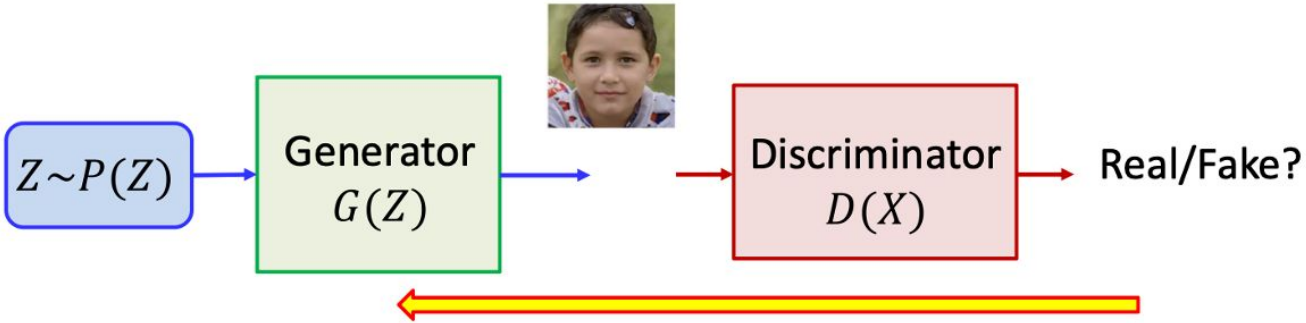
Training the discriminator



- **Training the discriminator:**
  - The discriminator is provided training examples of real and synthetic faces
  - The discriminator is trained to minimize its classification loss
    - Minimize error between actual and predicted labels
  - Discriminator parameters are trained such that
    - Maximize $\log(D(X))$ for real faces
    - Maximize $\log(1 - D(X))$ for synthetic faces

LECTURE RECAP

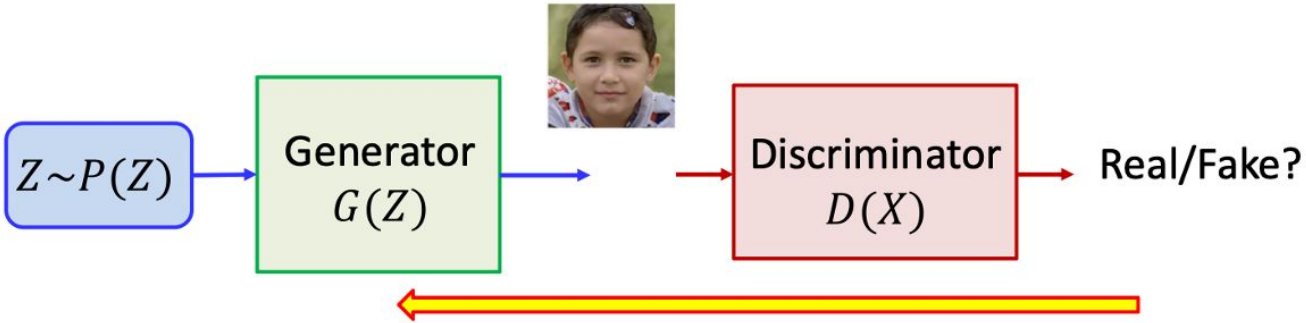
Training the generator



- **Training the generator:**
  - The discriminator's loss is backpropagated to the generator
  - The generator is trained to *maximize* the discriminator loss
    - It is trained to "fool" the discriminator
  - Generator parameters are trained such that
    - Minimize $\log\left(1 - D\big(G(Z)\big)\right)$

LECTURE RECAP

Training the generator



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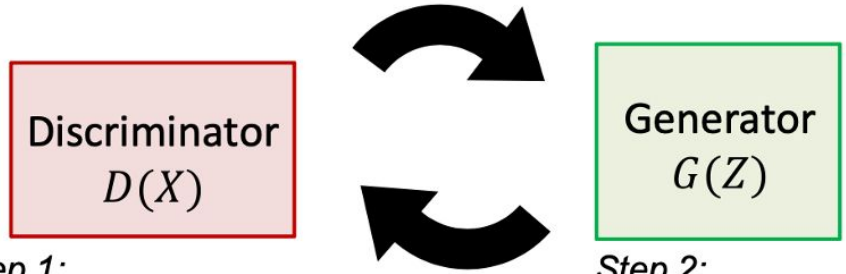
LECTURE RECAP

How to Train a GAN?



Step 1:
Train the Discriminator
using the current Generator

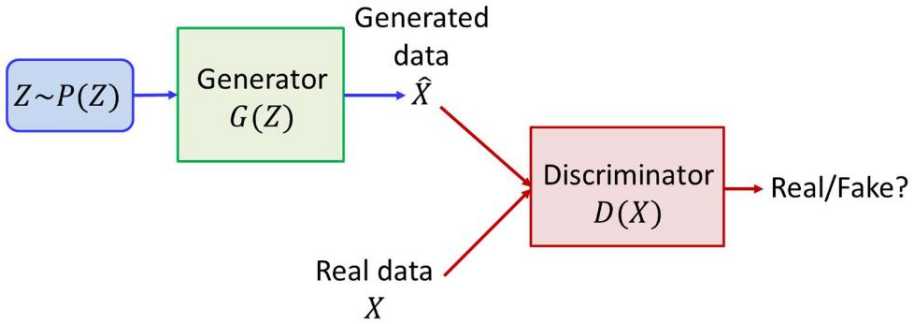
Step 2:
Train the Generator
to beat the Discriminator

Optimize: $\min_G \max_D E_X \log D(X) + E_Z \log(1 - D(G(Z)))$

The discriminator is not needed after convergence

LECTURE RECAP

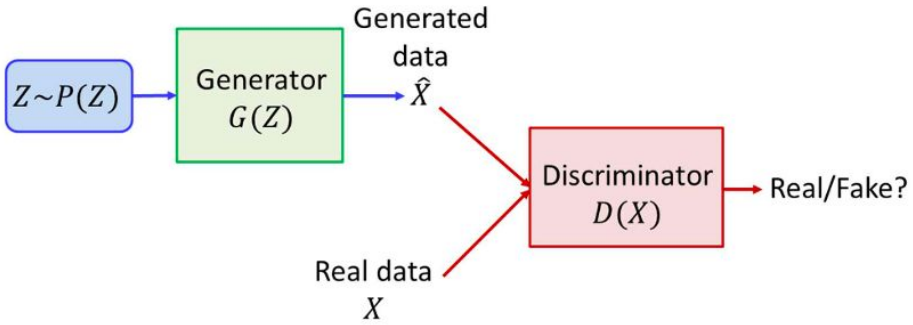
The GAN formulation



- Discriminator:
  - For real data X, Maximize $\log(D(X))$
  - For synthetic data Maximize $\log(1 - D(\hat{X}))$
- Generator
  - Minimize $\log(1 - D(\hat{X}))$

LECTURE RECAP

The GAN formulation



- The original GAN formulation is the following min-max optimization

$$\min_G \max_D \quad E_X \log D(X) + E_Z \log(1 - D(G(Z)))$$

- Objective of D: $D(X) = 1$ and $D(G(Z)) = 0$
- Objective of G: $D(G(Z)) = 1$

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5. **GAN Notebook**

Understanding Optimization Issues

Failures can be hard to diagnose

- “Mode collapse”
- Unclear errors

- Why? Many potential reasons

Key Questions about Training GANs

1. Why can’t we just train an optimal discriminator?

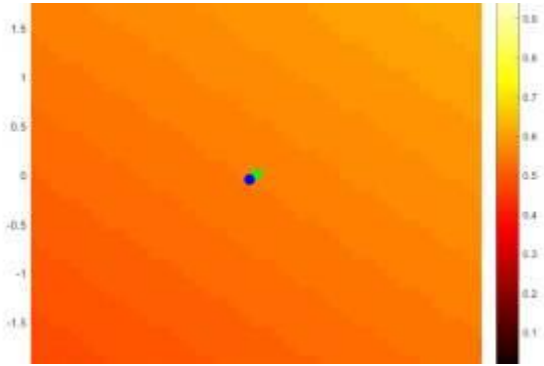
2. Which model is trained first?

3. Is one model trained more?

Simultaneous Updates

Why can’t we just train an optimal discriminator?

- The discriminator is conditional on the generator
- Too strong → no gradient for the generator
- Too weak → misleading signal



Adversarial Balance

- Both models must be trained together

Factors Affecting Balance

Who is ahead?

- Optimizers and learning rates
- Architectures, number of parameters, depths
- Regularization

Who is trained more?

- Repeat training D for k_D iterations, then train G for k_G iterations
- Make the iterations of k_D and k_G dynamic, depending on metrics

Not Guaranteed to Converge using Gradient Descent

Rock-Paper-Scissors example

- Player A randomly plays Rock
- Player B then should only play Paper
- Player A then should only play Scissors
- Player B then should only play Rock
- Player A then should only play Paper
- …

$$\mathbb{E}[W_A] = A_R B_S + A_P B_R + A_S B_P$$

Not Guaranteed to Converge using Gradient Descent

Rock-Paper-Scissors example

- Global Optimum:
  - both players choose with probability (0.33, 0.33, 0.33)

- Local Optimum for (rock, paper, scissors):
  - If player A plays with probability (0.36, 0.32, 0.32)
  - Then player B should play with probability (0,1,0)..

Not Guaranteed to reach a Stationary Point

- Gradients can circle or point away from the minima, so there might not be a local “well” around the stationary point.

- Even if a generated point is 0, it doesn’t mean that the points around that area area also 0; likewise, for 1.

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GAN Training Techniques

Foundations & Stability Fixes

- Gradient descent is locally stable
- Numerics of GANs
- Instance Noise
- Least-Squares GAN (LSGAN)
- DRAGAN
- Unrolled GAN
- Improved Techniques for GAN

Divergence Replacements

- Wasserstein GAN (WGAN)
- WGAN Gradient Penalty
- Spectral Normalized GAN

Least-Squares GAN

- Instead of binary cross entropy, uses an L2 loss

- The generator tries to minimize L2 loss $$\|a - D(G(Z))\|_2^2$$

- The discriminator tries to minimize L2 loss
$$\|b - D(G(Z))\|_2^2 + \|c - D(G(Z))\|_2^2$$

- Simplifies the loss function over time

Wasserstein GAN

- **Wasserstein Distance**: Mass times the distance required to transform one distribution to another.

- **Intuitive and Differentiable**: Provides smoother gradients compared to other metrics.

- **Hard to Calculate**: Involves optimal transport, which is computationally intensive.

Wasserstein GAN

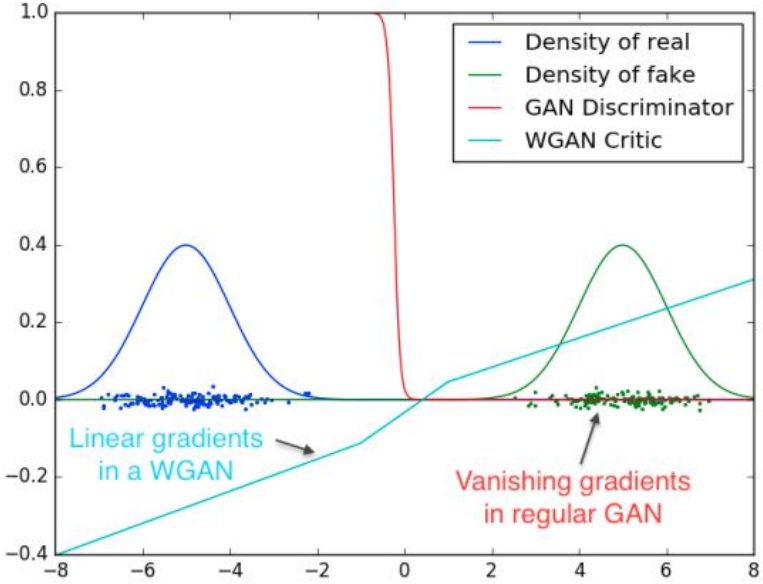

Figure 2: Optimal discriminator and critic when learning to differentiate two Gaussians. As we can see, the discriminator of a minimax GAN saturates and results in vanishing gradients. Our WGAN critic provides very clean gradients on all parts of the space.

Spectral Norm

- Max amount a matrix can stretch input vectors

$$\sup_x \frac{\|xA\|_2}{\|x\|_2} = \|A\|_2^*$$

- Divides weights by spectral norm to keep each layer stable

$$\hat{W} = \frac{W}{\|W\|_2^*}$$

- Helps keep discriminator smooth and well-behaved

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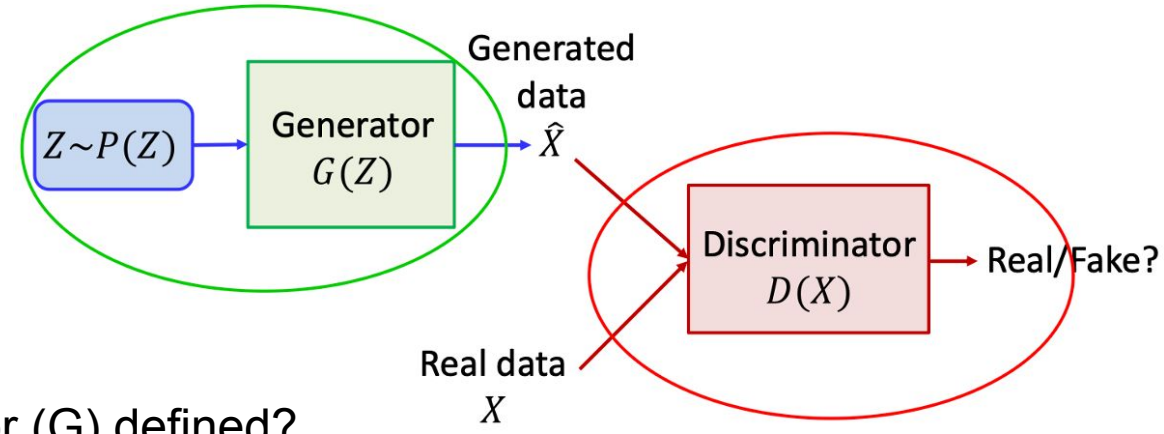
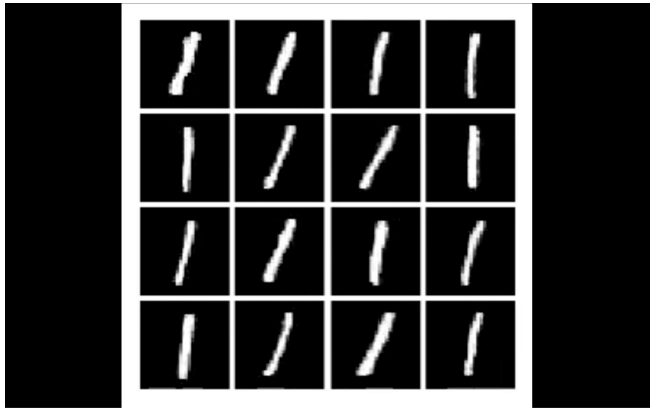
What to Look for in the Notebook



- Where is the Generator (G) defined?
- Where is the Discriminator (D) defined?
- Where does the Discriminator update happen?
- Where does the Generator update happen?
- How are real vs fake labels used?

What You Might Observe During Training

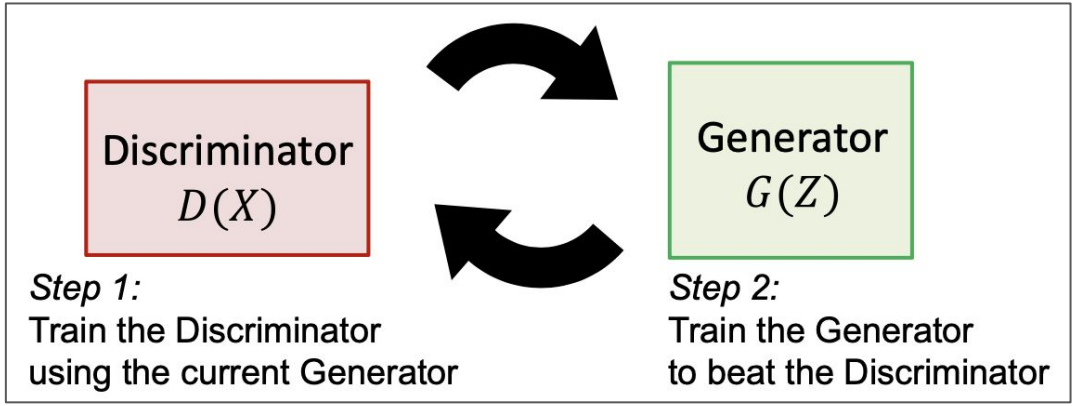
- Losses may oscillate instead of decreasing smoothly
- Generated images may repeat (mode collapse)
- The discriminator may become too strong or too weak
- Training may look unstable even when correct



How to Read the Training Loop

Step 1: Train Discriminator

- Real images → label = 1
- Fake images → label = 0

Step 2: Train Generator

- Generate fake images
- Label them as real
- Update G to fool D